Supplemental Online Content

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eTable 1. Countries for Which Guidelines for the Diagnostic Workup for Child Physical Abuse Were Specifically Searched on Website of the United Nations

eTable 2. Search Strategy

eFigure. Flowchart of Guidelines in the Systematic Review

eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse

eTable 4. Guidance Expected Given the Scope of the 20 Guidelines Included in the Systematic Review

eTable 5. Expected and Not Expected Statements Given the Scope of Guidelines for the Diagnosis of Child Abuse

eReferences

This supplemental material has been provided by the authors to give readers additional information about their work.

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eTable 1. Countries for Which Guidelines for the Diagnostic Workup for Child Physical Abuse Were Specifically Searched on Website of the United Nations

We restricted the research to countries with advanced economies according to the World Economic Situation and Prospects classification[1] and we added Israel.

Covered by the ECPCP[a]		Not covered by the ECPCP
Austria		Australia
Finland		Belgium
France		Canada
Germany		Denmark
Israel		Greece
Italy		Iceland
Luxembourg		Ireland
Portugal		Japan
Spain		Netherlands
Switzerland		New Zealand
		Norway
		Sweden
		United Kingdom
		United States

[a] European Confederation of Primary Care Paediatricians

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eTable 2. Search Strategy
(last update on June 2020)

a- Medline

1	(((((("Child Abuse"[Mesh] OR "child abuse[tiab]" OR "abused child[tiab]" OR "Battered Child Syndrome"[Mesh] OR "battered child syndrome[tiab]"OR "battered children[tiab]" OR "Wounds and Injuries"[Mesh] OR "non-accidental injury[tiab]") AND (Diagnosis[Mesh] OR diagnostic[tiab] OR diagnosis[tiab] OR investigation[tiab]) AND ("infant"[MeSH Terms] OR "child"[MeSH Terms] OR "adolescent"[MeSH Terms]) AND (Guideline[ptyp]) AND ("2000/01/01"[PDAT] : "2020/03/31"[PDAT]))))	114

b- Web of Science

1	TS: ((("child abuse" OR "abused child" OR "battered child syndrome" OR "battered children" OR "non-accidental injury") AND (diagnostic OR diagnosis OR investigation) AND (Guidelines OR recommendations))) OR TI: ((("child abuse" OR "abused child" OR "battered child syndrome" OR "battered children" OR "non-accidental injury") AND (diagnostic OR diagnosis OR investigation) AND (Guidelines OR recommendations))) Indexes=SCI-EXPANDED, SSCI, A&HCI, CPCI-S, CPCI-SSH, BKCI-S, BKCI-SSH, ESCI Timespan: 2000-2020	153

c- Google Scholar

The search strategy on Google Scholar was “physical abuse” “diagnosis” “detection” “guideline” and “child” -published since 2000-, limiting the analysis to the first 500 results.

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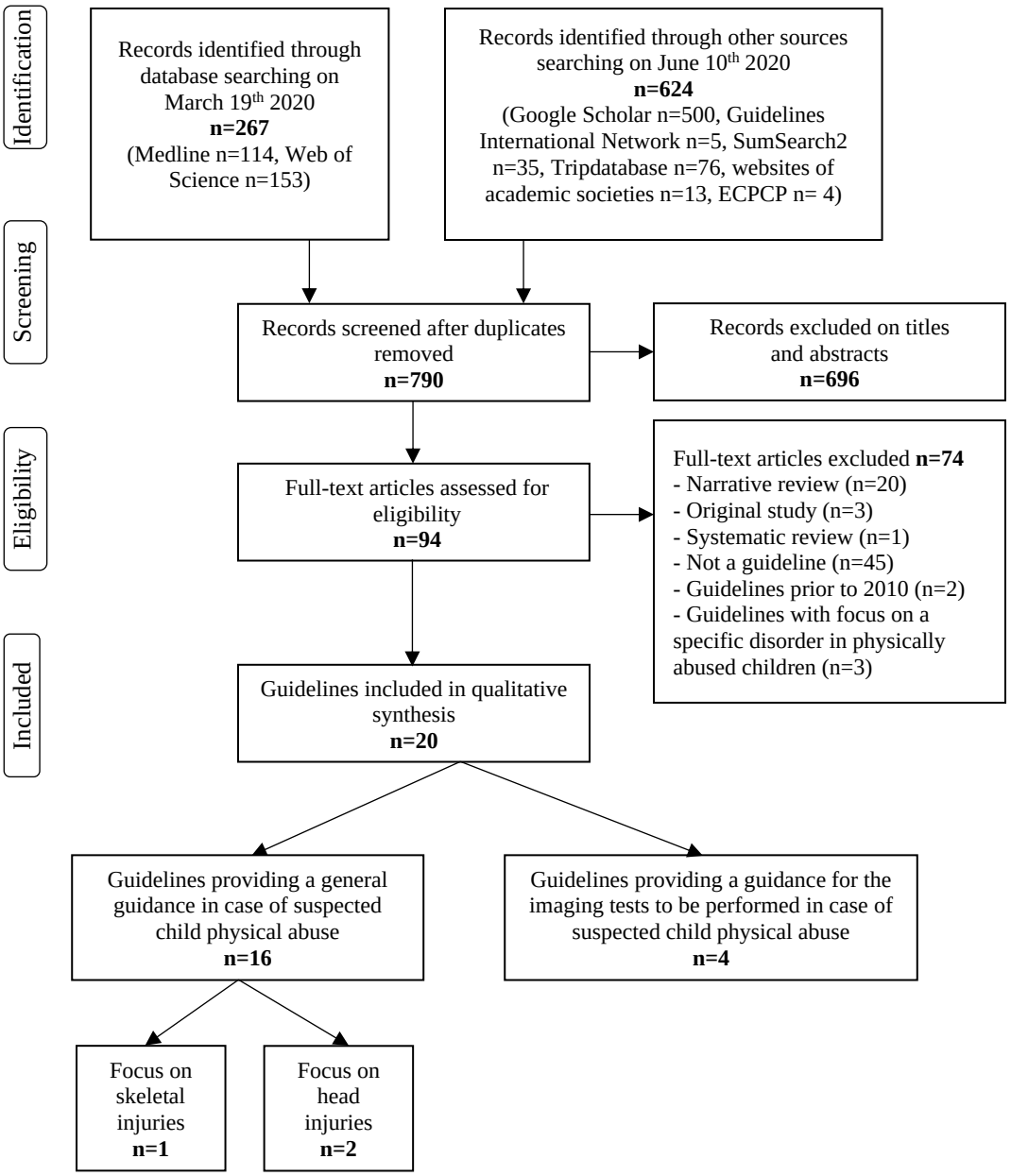
eFigure. Flowchart of Guidelines in the Systematic Review



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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse

Source	**New South Wales Government[2]**	**Government of Western Australia Department of Health[3]**	**Canadian Paediatric Society[4]**	**Haute Autorité de Santé[5]**	**Arbeitsgemeinschaft der Wissenschaftlichen Medizinischen Fachgesellschaften[6]**
Country	**Australia**	**Australia**	**Canada**	**France**	**Germany**
Year	**2014**	**2017**	**2018**	**2017**	**2019**
Objective for guidance	Physical abuse	Physical abuse	Inflicted skeletal injuries	Shaken baby syndrome	Physical abuse
	Definition of sentinel injuries				
Type	Yes	Yes	Yes	Yes	Yes
Location	NM	Yes	Yes	Yes	NM
Number	NM	Yes	NM	NM	NM
Size	NM	NM	NM	NM	NM
Pattern	NM	Yes	NM	NM	NM
According to child age and/or if cruising child	NM	Yes	Yes	Yes	NM
	Detection of skeletal or soft tissue injury				
Radiological skeletal survey	C	S <2 years	S <2 years	S <2 years	S <2 years
Number of views	NM	NM	24	19	17-19
Views	NM	NM	Frontal and lateral views of the skull, lateral views of the cervical spine and thoracolumbosacral spine, and single frontal views of the long bones, hands, feet, chest, and abdomen, oblique views of the ribs	S: frontal and lateral views of the skull (if no CT scan), frontal and lateral views of the spine, frontal view of the pelvis, frontal and lateral views of the chest, frontal views of each limb segment C: lateral views of the knees and ankles	Frontal and lateral views of the skull, frontal view of the chest, frontal views of the upper arms, forearms and hands, frontal views of the thighs, lower legs and feet If no rib fractures are detected: Lateral views of the chest If detection of one or more spinal fractures laterally, frontal view of the abdomen with pelvis and hips
Follow-up skeletal survey	NM	NM	C (14 days)	C (15 days)	C (11 to 14 days)
Number of views	NM	NM	10 minimum	13	16
Views	NM	NM	Omitting images of the skull, pelvis and lateral spine	S: frontal view of the pelvis, frontal and lateral views of the chest, frontal views of each limb segment C: lateral views of the knees and ankles	Frontal and lateral views of the skull, frontal view of the chest, frontal views of the upper arms, forearms and hands, frontal views of the thighs, lower legs and feet
Bone scintigraphy	C	C	C	C	C (ribs)
Sedation	NM	NM	NM	NM	NM
Ultrasonography (bones)	NM	NM	NM	NM	NM
Whole-body MRI	NM	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Sedation	NM	NM	NM	NM	NM
	Detection of head or spine injury				
Eye fundus examination	C	C <1 year	S	S	S < 2 years
Number, type, extent, patterns of retinal haemorrhages	NM	NM	Yes	NM	Yes

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	New South Wales Government[2]	Government of Western Australia Department of Health[3]	Canadian Paediatric Society[4]	Haute Autorité de Santé[5]	Arbeitsgemeinschaft der Wissenschaftlichen Medizinischen Fachgesellschaften[6]
Head CT	C	C	U	S <1 year	C (vital risk)
With contrast	NM	NM	NM	No	NM
Head MRI	C	C	U	S	C (no vital risk)
Abnormal CT scan	NM	NM	NM	Yes	Yes
Suspicion of abuse without neurological symptoms	NM	NM	NM	NM	NM
Sequences	NM	NM	NM	At least axial and sagittal T1 weighted images, axial T2 weighted images, diffusion weighted images and T2*	Diffusion weighted images
With contrast	NM	NM	NM	NM	NM
Sedation	NM	NM	NM	NM	NM
Spinal MRI	NM	NM	NM	C	C
Complete spine	NM	NM	NM	C	Yes
Cranial ultrasonography	NM	NM	NM	No	C
EEG	NM	NM	NM	S	NM
	Detection of thoracic or abdominal injury				
Chest CT	NM	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Abdominal ultrasonography	NM	U	U	C	NM
Abdominal CT	C	U	U	NM	NM
With contrast	NM	NM	NM	NM	NM
Pelvis CT	NM	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Laboratory tests					
Liver enzymes	C	C	S	S	S
Pancreatic enzymes	C	C	NM	S	S
Urinalysis	C (+toxicology)	C	S	NM	S
Renal function	NM	NM	S	NM	S
Troponin, creatine kinase	NM	C	NM	NM	S (creatine kinase)
	Differential diagnosis				
Calcium, phosphorus, alkaline phosphatase	C (+ magnesium)	NM	S	NM	S
25-hydroxyvitamin D, PTH	C	NM	C	NM	S
Skin biopsy for fibroblast culture and/or venous blood for DNA analysis for osteogenesis imperfecta	NM	NM	NM	NM	NM
CBC with platelets, coagulation regular (PT/INR/aPTT/fibrinogen)	C (PT, fibrinogen: NM)	C (coagulation: U)	S (coagulation: NM)	S (INR: NM; +lactates)	S (INR: NM)
Coagulation advanced (factor VIII level/factor IX level/VWF activity)	U	NM	NM	S (+factor XI)	C (+ factor XIII and blood group)
Serum copper, ceruloplasmin and vitamin C levels	NM	NM	C (vitamin C:NM)	NM	S
Urine organic acids[a]	C	NM	NM	C	C
New-born screen review	NM	NM	NM	NM	NM

CT, computed tomography; MRI, magnetic resonance imaging; EEG = electroencephalography
PTH, parathyroid hormone; CBC, complete blood cell count; PT, prothrombin time; INR, international normalized ratio; aPTT, activated partial thromboplastin time; VWF, von Willebrand factor
S, systematic; NM, not mentioned; C, case-by-case basis; U, unclear
[a] Performed for the detection of glutaric aciduria type 1

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	**Children's Health Ireland**[7]	**Association of Family Physicians**[8]	**Japan Pediatric Society**[9]	**Nederlandse Vereniging voor Kindergeneeskunde**[10]	**Starship / The Paediatric Society of New Zealand**[11]
Country	**Ireland**	**Israel**	**Japan**	**Netherland**	**New Zealand**
Year	**2020**	**2014**	**2014**	**2016**	**2016**
Objective for guidance	Physical abuse (imaging tests)	Physical abuse	Physical abuse	Physical abuse	Physical abuse
	Definition of sentinel injuries				
Type	NE	Yes	Yes	Yes	Yes
Location	NE	Yes	NM	Yes	Yes
Number	NE	NM	NM	Yes	NM
Size	NE	Yes	NM	Yes	NM
Pattern	NE	Yes	NM	Yes	Yes
According to child age and/or if cruising child	NE	NM	NM	Yes	Yes
	Detection of skeletal or soft tissues injury				
Radiological skeletal survey	S <2 years	S <36 months	S <2 years	S <2 years	C <2 years
Number of views	25	NM	17	21 to 24	NM
Views	Frontal and bilateral oblique views of the chest (3); abdomen and pelvis (1); frontal views of the upper and lower extremities, either each limb on a single view or more commonly two exposures for each limb (8); hands and feet (4); lateral spine (1); coned lateral views of the elbows, wrists, knees and ankles (8)	Spine, limbs and skull	Frontal view of the skull, frontal view of the thorax, lateral view of the cervical spine, lateral view of the lumbar spine, frontal view of the pelvis, frontal views of the upper and lower limbs, frontal view of the arms, frontal views of the forearm, frontal views of the hands, frontal views of the feet	Skull: AP and lateral Front suboccipital image, if clinically indicated Thorax: AP, including clavicles. Three quarters, front left and right front Abdomen: AP, including pelvis and hips Vertebral column: Lateral, entire spine (in multiple shots if needed). AP, if not imaged on thorax and abdomen. AP of the cervical spine (if indicated) Extremities: AP upper arms, AP forearms, AP thighs, AP lower legs, PA hands, AP feet.	NM
Follow-up skeletal survey	S (11 to 14 days)	NM	S (7 to 14 days)	C	C (14 days)
Number of views	11	NM	17	NM	NM
Views	This can be limited to frontal and bilateral oblique views of the chest, frontal views of the limbs and follow up of positive, suspected or equivocal findings on the primary skeletal survey. A repeat of the full skeletal survey may be undertaken if there is significant concern for occult injuries.	NM	Idem RSC → Skull front side, thorax (not chest) front side, cervical side, lumbar side, hip joint front, femur front (left and right) Front of lower leg (right and left) Front of humerus, front of forearm, front of phalange, front of toes	NM	NM
Bone scintigraphy	NM	C	NM	NM	C
Sedation	NM	NM	NM	NM	NM
Ultrasonography (bones)	NM	NM	NM	NM	NM
Whole-body MRI	NM	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Sedation	NM	NM	NM	NM	NM
	Detection of head or spine injury				
Eye fundus examination	NE	S < 2 years	S	C	C <2 years
Number, type, extent, patterns of retinal haemorrhages	NE	NM	Yes	NM	NM

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	Children's Health Ireland[7]	Association of Family Physicians[8]	Japan Pediatric Society[9]	Nederlandse Vereniging voor Kindergeneeskunde[10]	Starship / The Paediatric Society of New Zealand[11]
Head CT	S <1 year C >1 year	C	S < 2 years (or MRI)	C	S <1 year C >1 year
With contrast	No	No	NM	NM	No
Head MRI	C	NM	S (or CT)	C	C
Abnormal CT scan	Yes	NM	Yes	Yes	Yes
Suspicion of abuse without neurological symptoms	NM	NM	NM	Yes	Yes
Sequences	NM	NM	Including diffusion weighted images	NM	Gradient echo sequences, diffusion-weighted images and susceptibility-weighted images
With contrast	NM	NM	NM	NM	NM
Sedation	NM	NM	U	NM	NM
Spinal MRI	C	NM	NM	C	C
Complete spine	Yes	NM	NM	Yes	Yes
Cranial ultrasonography	NM	NM	NM	NM	NM
EEG	NE	NM	NM	NM	NM
	Detection of thoracic or abdominal injury				
Chest CT	C	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Abdominal ultrasonography	U	NM	NM	NM	NM
Abdominal CT	U	C	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Pelvis CT	NM	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Laboratory tests					
Liver enzymes	NE	C	NM	NM	NM
Pancreatic enzymes	NE	C	NM	NM	NM
Renal function	NE	NM	NM	NM	NM
Urinalysis	NE	C (+toxicology)	NM	NM	NM
Troponin, creatine kinase	NE	NM	NM	NM	NM
	Differential diagnosis				
Calcium, phosphorus, alkaline phosphatase	NE	NM	NM	NM	NM
25-hydroxyvitamin D, PTH level	NE	NM	NM	NM	NM
Skin biopsy for fibroblast culture and/or venous blood for DNA analysis for osteogenesis imperfecta	NE	NM	NM	NM	NM
CBC with platelets, coagulation regular (PT/INR/aPTT/fibrinogen)	NE	C (coagulation: U)	NM	C (INR: NM)	C (PT, fibrinogen: NM)
Coagulation advanced (factor VIII level/factor IX level/VWF activity)	NE	NM	NM	C (+ factor XIII)	C (VWF activity + blood group)
Serum copper, ceruloplasmin and vitamin C levels	NE	NM	NM	C (vitamin C)	NM
Urine organic acids[a]	NE	NM	NM	NM	C
New-born screen review	NE	NM	NM	NM	NM

CT, computed tomography; MRI, magnetic resonance imaging; EEG = electroencephalography
PTH, parathyroid hormone; CBC, complete blood cell count; PT, prothrombin time; INR, international normalized ratio; aPTT, activated partial thromboplastin time; VWF, von Willebrand factor
S, systematic; NM, not mentioned; C, case-by-case-basis; NE, not expected; U, unclear
[a] Performed for the detection of glutaric aciduria type 1

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	Nasjonalt kunnskapassenter om vold og traumatisk stress[12]	Asociación Española de Pediatría and Sociedad Española de Urgencias Pediátricas[13]	Swedish Pediatric Society[14]	Swedish Paediatric Radiology Society[15]	Swiss Society of Paediatrics[21]
Country	**Norway**	**Spain**	**Sweden**	**Sweden**	**Switzerland**
Year	**2018**	**2010**	**2019**	**2019**	**2017**
Objective for guidance	Physical abuse	Physical abuse	Physical abuse	Physical abuse (imaging tests)	Physical abuse
	Definition of sentinel injuries				
Type	Yes	Yes	Yes	NE	Yes
Location	Yes	NM	NM	NE	NM
Number	No	Yes	NM	NE	NM
Size	No	NM	NM	NE	NM
Pattern	No	NM	NM	NE	NM
According to child age and/or if cruising child	No	NM	NM	NE	NM
	Detection of skeletal or soft tissues injury				
Radiological skeletal survey	S <2 years	S <2 years	S	S <2 years	C <2 years
Number of views	22	NM	NM	28-32	NM
Views	Skull - front, sides (both sides towards detector), possibly Townes projection; should be performed in anyone under two years of age (even if cerebral CT has been performed) Thorax skeleton, front, side, oblique (skeletal exposure) Whole column (C-Th-LS), side Abdomen including pelvis, front Upper and forearms (AP), hands (PA) Thighs, calves and feet (AP)	NM	NM	Frontal and lateral views of the skull, frontal view and lateral views of the chest, frontal view of the abdomen and pelvis, lateral view of the spine, frontal view of the upper arms, forearms, elbows and hand, frontal view of the upper and lower limbs, knees, ankles, feet	NM
Follow-up skeletal survey	C (11-14 days)	NM	S (14 days)	S (11-14 days)	NM
Number of views	NM	NM	NM	NM	NM
Views	NM	NM	Whole-body skeleton X-ray	NM	NM
Bone scintigraphy	No	C	NM	NM	C > 2 years
Sedation	NM	NM	NM	NM	NM
Ultrasonography (bones)	NM	NM	NM	NM	NM
Whole-body MRI	No	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Sedation	NM	NM	NM	NM	NM
	Detection of head or spine injury				
Eye fundus examination	C <5 years old	S	S	NE	S
Number, type, extent, patterns of retinal haemorrhages	Yes	NM	Yes	NE	NM
Head CT	S <1 year, C 1 to 2 years old	S	S	S <1 year C >1 year	S
With contrast	No	NM	NM	No	NM
Head MRI	C	C	C	C	S
Abnormal CT scan	Yes	NM	Yes	Yes	NM
Suspicion of abuse without neurological symptoms	No	NM	Yes	NM	NM

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	Nasjonalt kunnskapassenter om vold og traumatisk stress[12]	Asociación Española de Pediatría and Sociedad Española de Urgencias Pediátricas[13]	Swedish Pediatric Society[14]	Swedish Paediatric Radiology Society[15]	Swiss Society of Paediatrics[21]
Sequences	Head: Axial T2, coronal T2, coronal and axial FLAIR, T1 volume series, susceptibility weighted series, diffusion weighted imaging. Contrast-enhanced T1 volume series can be considered. Spinal canal: Sagittal T1, STIR and T2 weighted recordings, possibly axial recordings if visible pathology on the sagittal (axial T1 and T2 over the relevant area).	NM	NM	T1 sagittal and axial T2 axial T2 flair coronal Susceptibility weighted imaging or equivalent / diffusion	NM
With contrast	Yes (C)	NM	NM	NM	NM
Sedation	Yes	NM	NM	NM	NM
Spinal MRI	Yes	NM	C	C	NM
Complete spine	No	NM	Yes	Yes	NM
Cranial ultrasonography	NM	C	NM	NM	NM
EEG	NM	NM	C	NE	NM
	Detection of thoracic or abdominal injury				
Chest CT	C	NM	C	C	NM
With contrast	NM	NM	NM	NM	NM
Abdominal ultrasonography	NM	NM	NM	NM	C
Abdominal CT	C	NM	C	C	C
With contrast	Yes	NM	NM	NM	NM
Pelvis CT	NM	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Laboratory tests					
Liver enzymes	C	C	S	NE	C
Pancreatic enzymes	C	NM	NM	NE	C
Renal function	C	NM	S	NE	C
Urinalysis	C	NM	S	NE	C
Troponin, creatine kinase	C (creatine kinase)	NM	S (troponin)	NE	NM
	Differential diagnosis				
Calcium, phosphorus, alkaline phosphatase	C (+ Na)	NM	S (+ K, Na)	NE	C (phosphorus, alkaline phosphatase: NM, + Na, K, Cl)
25-hydroxyvitamin D, PTH level	C	NM	S	NE	NM
Skin biopsy for fibroblast culture and/or venous blood for DNA analysis for osteogenesis imperfecta	C	NM	NM	NE	NM
CBC with platelets, coagulation regular (PT/INR/aPTT/fibrinogen)	C (PT: NM)C	S (coagulation: U)	S (+lactates)	NE	C (aPTT: NM)
Coagulation advanced (factor VIII level/factor IX level/VWF activity)	C (+ factor VII + vitamin K)	NM	S	NE	C (VWF activity, + factor XIII)
Serum copper, ceruloplasmin and vitamin C levels	C (vitamin C: NM)	NM	S (vitamin C: NM)	NE	NM
Urine organic acids*	C	NM	NM	NE	C
New-born screen review	NM	NM	NM	NE	C

CT, computed tomography; MRI, magnetic resonance imaging; EEG = electroencephalography
PTH, parathyroid hormone; CBC, complete blood cell count; PT, prothrombin time; INR, international normalized ratio; aPTT, activated partial thromboplastin time; VWF, von Willebrand factor ; K, Potassium, Na, Sodium, Cl, Chloride.
S, systematic; NM, not mentioned; C, case-by-case-basis; NE, not expected; U, unclear
[a] Performed for the detection of glutaric aciduria type 1

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	The Royal College of Radiologists and The Society & College of Radiographers[16]	Royal College of Paediatrics and Child Health[17]	American Academy of Pediatrics[18]	American College of Radiology[19]	American Academy of Pediatrics[20]
Country	**UK**	**UK**	**USA**	**USA**	**USA**
Year	**2018**	**2019**	**2015**	**2017**	**2020**
Objective for guidance	Physical abuse (imaging tests)	Physical abuse	Physical abuse	Physical abuse (imaging tests)	Abusive head trauma
	Definition of sentinel injuries				
Type	NE	Yes	Yes	NE	Yes
Location	NE	Yes	Yes	NE	Yes
Number	NE	Yes	Yes	NE	NM
Size	NE	No	NM	NE	NM
Pattern	NE	Yes	Yes	NE	NM
According to child age and/or if cruising child	NE	Yes	Yes	NE	NM
	Detection of skeletal or soft tissues injury				
Radiological skeletal survey	S <2 years	S <2 years	S <2 years	S <2 years	S <2 years
Number of views	24-25	24-25	24	24	NM
Views	Frontal and lateral views of the skull, frontal view and lateral views of the chest, frontal view of the abdomen and pelvis, lateral views of the spine, frontal view of the arms (or separate humerus and separate forearm), lateral views of the elbows, wrists, frontal views of the hands and wrists, frontal views of the upper and lower limbs (or separate femur or separate tibia/fibula)	Frontal and lateral views of the skull, frontal view and lateral views of the chest, frontal view of the abdomen and pelvis, lateral views of the spine, frontal view of the arms (or separate humerus and separate forearm), lateral views of the elbows, wrists, frontal views of the hands and wrists, frontal views of the upper and lower limbs (or separate femur or separate tibia/fibula)	Frontal and lateral views of the skull, lateral views of the cervical spine and thoracolumbosacral spine, and single frontal views of the long bones, hands, feet, chest, and abdomen, oblique views of the ribs	Frontal and lateral views of the skull, lateral views of the cervical spine and thoracolumbosacral spine, and single frontal views of the long bones, hands, feet, chest, and abdomen, oblique views of the ribs	NM
Follow-up skeletal survey	S (11 to 14 days, max 28 days)	S (11 to 14 days)	C (14 to 21 days)	C (14 days)	NM
Number of views	9 minimum	NM	NM	10 minimum	NM
Views	Any abnormal or suspicious areas + frontal and lateral views of the chest, frontal views of the arms, frontal views of the upper and lower limbs	NM	NM	To limit radiation exposure, pelvis, spine, and skull radiographs can be omitted if no injury was initially seen in these regions	NM
Bone scintigraphy	No	C	C	C	NM
Sedation	NM	NM	NM	Yes	NM
Ultrasonography (bones)	C	NM	NM	NM	NM
Whole-body MRI	No	NM	NM	NM	NM
With contrast	NM	NM	NM	NM	NM
Sedation	NM	NM	NM	NM	NM
	Detection of head or spine injury				
Eye fundus examination	NE	S	S	NE	U
Number, type, extent, patterns of retinal haemorrhages	NE	Yes	Yes	NE	S
Head CT	S	S <1year, C 1 to 2years old	C	S	S (or MRI)
With contrast	NM	NM	Yes	No	NM
Head MRI	C	C	C	C	C
Abnormal CT scan	Yes	Yes	Yes	Yes	S (or MRI)

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Suspicion of abuse without neurological symptoms	Yes	NM	Yes	Yes	NM

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eTable 3. Recommended Imaging and Laboratory Tests for Suspicion of Physical Abuse (continued)

Source	The Royal College of Radiologists and The Society & College of Radiographers[16]	Royal College of Paediatrics and Child Health[17]	American Academy of Pediatrics[18]	American College of Radiology[19]	American Academy of Pediatrics[20]
Sequences	T1 sagittal and axial T2 axial Axial or coronal fluid-attenuated inversion recovery (FLAIR) Axial T2 gradient echo or susceptibility-weighted imaging / diffusion weighted imaging	Diffusion weighted imaging	Diffusion-weighted images	Standard sequences + diffusion weighted imaging and susceptibility weighted imaging +/- contrast-enhanced MRI sequences	NM
With contrast	NM	NM	NM	No	NM
Sedation	NM	NM	NM	Yes	NM
Spinal MRI	C	C	C	S	C
Complete spine	Yes	Yes	U	Yes	U
Cranial ultrasonography	NM	No	C	NM	No
EEG	NE	NM	NM	NE	NM
	Detection of thoracic or abdominal injury				
Chest CT	C	NM	NM	C	NM
With contrast	NM	NM	NM	C	NM
Abdominal ultrasonography	NM	NM	NM	No	NM
Abdominal CT	S	C	C	S	NM
With contrast	NM	Yes	Yes	Yes	NM
Pelvis CT	NM	NM	NM	S	NM
With contrast	NM	NM	NM	Yes	NM
Laboratory tests					
Liver enzymes	NE	C	S	NE	NM
Pancreatic enzymes	NE	C	S	NE	NM
Renal function	NE	NM	NM	NE	NM
Urinalysis	NE	NM	C	NE	NM
Troponin, creatine kinase	NE	NM	S	NE	NM
	Differential diagnosis				
Calcium, phosphorus, alkaline phosphatase	NE	C	S	NE	NM
25-hydroxyvitamin D, PTH level	NE	C	C	NE	NM
Skin biopsy for fibroblast culture and/or venous blood for DNA analysis for osteogenesis imperfecta	NE	C	C	NE	NM
CBC with platelets, coagulation regular (PT/INR/aPTT/fibrinogen)	NE	C	S	NE	NM
Coagulation advanced (factor VIII level/factor IX level/VWF activity)	NE	C (factor IX, d-dimer: NM; + factor XIII)	S (+ D-dimer)	NE	NM
Serum copper, ceruloplasmin and vitamin C levels	NE	U (copper, vitamin C) + vitamin A	C	NE	NM
Urine organic acids*	NE	C	C	NE	NM
New-born screen review	NE	NM	S	NE	NM

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PTH, parathyroid hormone; CBC, complete blood cell count; PT, prothrombin time; INR, international normalized ratio; aPTT, activated partial thromboplastin time; VWF, von Willebrand factor
S, systematic; NM, not mentioned; C, case-by-case basis; NE, not expected; U, unclear
[a] Performed for the detection of glutaric aciduria type 1

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eTable 4. Guidance Expected Given the Scope of the 20 Guidelines Included in the Systematic Review

			Specific guidance expected				
Country (year)	**Title (verbatim +/- translation)**	**Scope (verbatim +/- translation)**	**Sentinel injury**	**Skeletal injury**	**Head and spine injury**	**Thoracic or abdominal injury**	**Differential diagnosis**
Australia (2014)[2]	"Suspected child abuse and neglect medical protocol"	"This guideline provides a new standard template for medical staff to record a forensically oriented medical assessment of a child or young person, to enable an informed opinion about the probability that injuries have been caused intentionally."	Expected	Expected	Expected	Expected	Expected
Australia (2017)[3]	"Guidelines for protecting children 2015"	"The guidelines for protecting children […] provide information, procedural guidance and resources to support WA health system staff to promote and safeguard the wellbeing of children when they are at risk of or have been harmed through abuse and/or neglect."	Expected	Expected	Expected	Expected	Expected
Canada (2018)[4]	"The medical assessment of fractures in suspected child maltreatment: infants and young children with skeletal injury"	-	Expected	Expected	Expected	Expected	Expected
France (2017)[5]	"Shaken baby syndrome or non-accidental shaking head injury"	"The objectives are to improve the recognition of the shaken baby syndrome by health professionals and to specify the approach and diagnostic criteria; to specify the possible mechanisms of injury; to specify the action to be taken to protect the infant."	Expected	Expected	Expected	Expected	Expected
Germany (2019)[6]	"Child maltreatment, abuse, neglect with the involvement of youth welfare and pedagogy (child protection guideline)"	-	Expected	Expected	Expected	Expected	Expected
Ireland (2020)[7]	"Guidelines for investigation of children with suspected physical abuse"	"To provide clinical guidelines for the radiological investigation of children with suspected physical abuse"	Not expected	Expected	Expected (except eye fundus examination)	Expected	Not expected
Israel (2014)[8]	"Identification and treatment in minors' victims of abuse and neglect"	"This guideline regulates the modalities of treatment by the health services in the hospital and in the community, in all cases where there is a suspicion of harm and abuse of the juvenile by the person in charge."	Expected	Expected	Expected	Expected	Expected
Japan (2014)[9]	"Child abuse treatment guide"	-	Expected	Expected	Expected	Expected	Expected
Netherland (2016)[10]	"Guideline on identifying child abuse in emergency medical care"	"The directive focuses on all children presenting in emergency medical care. The described next steps are exclusively for those children who are suspected of child abuse."	Expected	Expected	Expected	Expected	Expected
New Zealand (2016)[11]	"Abuse and neglect"	"This guideline is intended to help you in the appropriate assessment and initial management of children and young people where you become concerned about possible abuse or neglect."	Expected	Expected	Expected	Expected	Expected
Norway (2018)[12]	"Investigation of children on suspicion of violence and abuse"	-	Expected	Expected	Expected	Expected	Expected
Spain (2010)[13]	"Child abuse. Intervention in the emergencies"	-	Expected	Expected	Expected	Expected	Expected
Sweden (A) (2019)[14]	"Medical investigation and collaboration in case of suspicion of abuse of young children"	-	Expected	Expected	Expected	Expected	Expected
Sweden (B) (2019)[15]	"Guidelines for radiology in the investigation of suspected physical child abuse"	-	Not expected	Expected	Expected (except eye fundus examination)	Expected	Not expected
Switzerland (2017)[21]	"Guidelines for child protection activities in paediatric hospitals"	"[...] guidelines for the diagnosis and management of child abuse and neglect."	Expected	Expected	Expected	Expected	Expected
UK (2018)[16]	"The radiological investigation of suspected physical abuse in children"	"This guidance is designated to assist referring clinicians, paediatricians, radiologists, radiographers and nuclear medicine technologists who are requesting, performing or reporting on imaging in such cases […]"	Not expected	Expected	Expected (except eye fundus examination)	Expected	Not expected
UK (2019)[17]	"Recognition of physical abuse. Good Practice recommendations"	"This chapter provides a guide to the features of childhood injury that should raise suspicion of physical abuse and the assessment and investigations that should be performed to determine the likelihood of physical abuse."	Expected	Expected	Expected	Expected	Expected
USA (2015)[18]	"The evaluation of suspected child physical abuse"	-	Expected	Expected	Expected	Expected	Expected
USA (2017)[19]	ACR Appropriateness Criteria® Suspected Physical Abuse—Child	"These criteria are intended to guide radiologists, radiation oncologists, and referring physicians in making decisions regarding radiologic imaging and treatment."	Not expected	Expected	Expected (except eye	Expected	Not expected

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					fundus examination)		
USA (2020)[20]	Abusive head trauma in infants and children	"This statement aims to provide pediatric practitioners with general guidance on a complex subject."	Expected	Expected	Expected	Expected	Expected

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eTable 5. Expected and Not Expected Statements Given the Scope of Guidelines for the Diagnosis of Child Abuse

Expected statement[a]	Guideline scope	
	General guidance including imaging and laboratory tests	**Guidance for the imaging tests to be performed**
Radiological skeletal survey	Expected	Expected
Follow-up skeletal survey	Expected	Expected
Bone scintigraphy	Expected	Expected
Eye fundus examination	Expected	Not expected
Head CT	Expected	Expected
Head MRI	Expected	Expected
Spinal MRI	Expected	Expected
Cranial ultrasonography	Expected	Expected
Chest CT	Expected	Expected
Abdominal ultrasonography	Expected	Expected
Abdominal CT	Expected	Expected
Liver enzymes	Expected	Not expected
Pancreatic enzymes	Expected	Not expected
Renal function	Expected	Not expected
Urinalysis	Expected	Not expected
Troponin, creatine kinase	Expected	Not expected
Calcium, phosphorus, alkaline phosphatase	Expected	Not expected
25-hydroxyvitamin D, PTH	Expected	Not expected
Skin biopsy for fibroblast culture and/or venous blood for DNA analysis for osteogenesis imperfecta	Expected	Not expected
Serum copper, ceruloplasmin and vitamin C levels	Expected	Not expected
Urine organic acids	Expected	Not expected
CBC with platelets, coagulation regular (PT/INR/aPTT/fibrinogen)	Expected	Not expected
Coagulation advanced (factor VIII level/factor IX level/d-dimer/VWF activity)	Expected	Not expected

CT, computed tomography; MRI, magnetic resonance imaging; US, ultrasonography
PTH, parathyroid hormone; CBC, complete blood cell count; PT, prothrombin time; INR, international normalized ratio; aPTT, activated partial thromboplastin time; VWF, von Willebrand factor

[a] Suggested in >2 of the 20 included guidelines

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